

Time Resolved Absorption Spectroscopy of Zinc-Tetraphenylporphyrin and Quenched Samples

Sara Wavde

Department of Physics and Astronomy, University of Manchester

(Experiment performed in collaboration with Finley Groom)

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The objective of this experiment was to measure the decay times of excited zinc-tetraphenylporphyrin (ZnTPP) molecules using time-resolved absorption spectroscopy and to determine the quenching rate constants. This was achieved by employing pump and probe lasers in the nanosecond range to excite the porphyrin molecules. The bimolecular quenching rate constants obtained were $k_q = (1.17 \pm 0.03) \times 10^{-4} \text{ L mmol}^{-1} \text{ ns}^{-1}$ for self-quenching, $k_q = (1.00 \pm 0.05) \times 10^{-3} \text{ L mmol}^{-1} \text{ ns}^{-1}$ for C_{70} , and $k_q = (7.3 \pm 0.4) \times 10^{-4} \text{ L mmol}^{-1} \text{ ns}^{-1}$ for C_{60} . The C_{70} quenching rate constant shows good agreement with the literature value of $9.8 \times 10^{-4} \text{ L mmol}^{-1} \text{ ns}^{-1}$ [1], with the other values also following the trends outlined in the literature.

1. INTRODUCTION

The behaviour of molecules following photoexcitation is fundamental to various phenomena, ranging from photosynthesis to photodynamic therapy (PDT). When a molecule absorbs light, it transitions into an electronically excited state before decaying by either fluorescence, phosphorescence, or non-radiative pathways. In this experiment, the transient absorption in the nanosecond range was explored using a pump-probe laser setup. In particular, the nanosecond range bridges the gap between ultrafast processes and slower diffusive processes. Using a pump laser to excite the molecules and a probe laser to investigate the resulting effects, the relaxation of the molecules was studied. In addition, quenchers are of interest because they provide a means to study bimolecular interactions and energy transfer processes. When a quencher molecule collides with an excited porphyrin, it provides an additional non-radiative decay pathway, reducing the observed excited-state lifetime and emission intensity. Thus, this experiment studied the influence of intermolecular interactions in photoexcited systems.

2. THEORY AND METHODOLOGY

2.1. Excitation and Relaxation Mechanisms

When there is a photoinduced excitation of ZnTPP, the pump laser excites it into a singlet state, which then undergoes intersystem crossing to populate the triplet state. This process is outlined in the Jablonski diagram below.

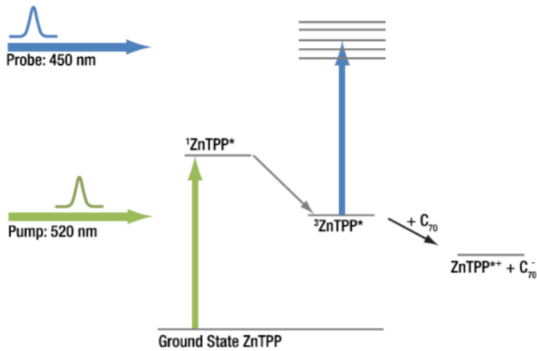


FIG. 1. A Jablonski diagram showing the excitation and de-activation paths for the TA processes in ZnTPP and C_{70} [2].

The Transient Absorption (TA) is the difference in absorbance before and after the excitation denoted as ΔA , and is related to intensity by

$$\Delta A = -\log \frac{I_{\text{pump}}}{I_{\text{no pump}}}, \quad (1)$$

where I_{pump} and $I_{\text{no pump}}$ are the intensities of the probe lasers with or without pumps respectively. This excited state is dependent on the concentration of excited ZnTPP molecules and reaction rates, where

$$-\frac{dC_T(t)}{dt} = k_1 C_T(t) + k_2 (C_T(t))^2 + k_3 C_T(t) C_G(t). \quad (2)$$

The first term on the right hand side represents the first order decay from the triplet state to the ground state, including both radiative (phosphorescence) and non-radiative (internal conversion) pathways. The second and third terms are bimolecular deactivation mechanisms, where excited metastable states can take part in deactivation through triplet-triplet annihilation or triplet-ground state quenching [3]. These processes are shown in Figure 1. By assuming that $C_G(t)$ does not change significantly and that $(C_T(t))^2 \ll C_T(t) C_G(t)$, the differential equation simplifies to

$$-\frac{dC_T(t)}{dt} = k_0 C_T(t), \quad (3)$$

in which $k_0 = k_1 + k_3 C_G$. Solving this differential equation gives

$$C_T(t) = C_T(0) e^{-k_0 t}, \quad (4)$$

which is the exponential decay observed experimentally. The decay time is therefore $\tau = 1/k_0$.

2.2. Time-Resolved Absorption Spectroscopy (TRAS)

To experimentally observe the exponential relation seen in the equation, the pump-probe technique used in this experiment is a form of time-resolved absorption spectroscopy. A pump laser pulse excites the sample as seen in the Jablonski diagram, creating an excited state population. A delayed probe pulse measures the transmission of the sample as a function of the time delay between the two pulses. The change in absorbance, $\Delta A(t)$, is proportional to the concentration of excited states, allowing the decay kinetics to be monitored.

2.3. Quenching Kinetics and the Stern–Volmer Equation

The excited state can decay via intrinsic pathways (fluorescence, non-radiative decay) with rate constant k_0 . In the presence of a quencher Q, an additional bimolecular quenching pathway exists:



The observed decay rate becomes:

$$k_{\text{obs}} = k_0 + k_q[Q], \quad (6)$$

where k_q is the bimolecular quenching rate constant [4]. This equation shows that as the quencher concentration increases, the observed triplet decay rate increases, signifying that the triplet state returns to the ground state more quickly through collisional energy transfer to the quencher.

2.4. Experimental Setup

The setup consists of two pulsed lasers triggered with a fixed delay via a Data Acquisition (DAQ) card, controlled with ThorLabs LabVIEW software [2]. For every pump pulse, two probe pulses are fired, with the beams reflected by mirrors to overlap at the sample. An iris blocks the pump beam so only the probe reaches the photodiode detector. The transient absorption signal is obtained from the difference in probe intensity measured with and without the pump.

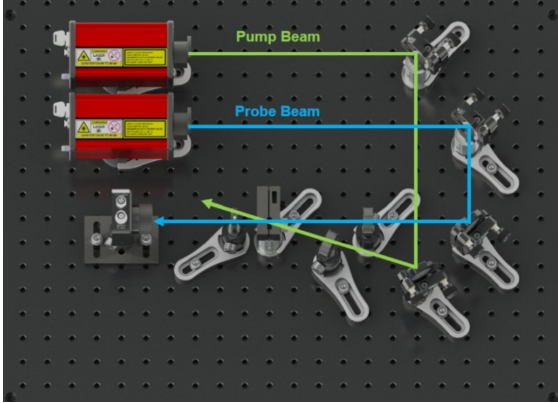


FIG. 2. Diagram of the TRAS experimental setup. The probe laser (blue path) and pump laser (green path) are focused on the sample. An iris blocks the pump beam after the sample so only the probe reaches the detector. [2]

The pulses shine on the samples in cuvettes, which are made from base chemicals ZnTPP or fullerene (C_{70} or C_{60}), both using benzonitrile solvents. A small mass m for each base chemical is weighed using a sensitive scale, and the required volume V of the solvent is calculated by

$$V = \frac{m}{C \cdot M}, \quad (7)$$

where C is the desired concentration and M are the molecular weights of the chemicals with constant values for each base [2]. The samples are made by mixing the volumes from the base solution, often using a sonicator to ensure a thorough mixture. By doing so, different concentrations of the base solutions can be obtained. In this experiment, the following samples were prepared:

- Six concentrations of ZnTPP ranging from 0.8 mmol/L (BZ08) to 0.1 mmol/L (BZ01).
- Five concentrations of C_{70} samples ranging from 1.0 mmol/L (BZC1) to 0.1 mmol/L (BZC01).
- Four concentrations of C_{60} samples ranging from 1.0 mmol/L (BZbb1) to 0.25 mmol/L (BZbb025).

By following the experimental procedure and repeating the measurements for each concentration, the Stern–Volmer plot was obtained.

2.5. Comparison of C_{70} and C_{60} as Quenchers

C_{70} and C_{60} were used as quenchers for this experiment due to their differing photophysical properties. They both follow the fullerene structure but differ in its symmetry: C_{60} has a spherical football-like shape, while C_{70} has a more elongated rugby-ball shape [5]. C_{70} has a significantly longer triplet excited state lifetime and stronger visible light absorption compared to C_{60} , meaning that it remains available for bimolecular collisions for longer. By comparing the quenching behaviour of both fullerenes, the dependence of triplet state quenching kinetics can be explored.

3. RESULTS AND DISCUSSION

The apparent quenching rate constants k_{app} were obtained for three sample types: self-quenching of ZnTPP, quenching by C_{70} , and quenching by C_{60} . For each sample set, k_{app} was plotted against the quencher concentration and a linear fit was performed, where the gradient yields the bimolecular quenching constant k_q . To obtain robust uncertainty estimates on the fitted decay times, a residual bootstrapping procedure was used. The best-fit model was first obtained by least-squares optimisation, after which the fit residuals were resampled with replacement 5000 times. Each resampled residual set was added back to the best-fit curve to generate a synthetic dataset, which was then refitted using the same model and initial parameters, yielding a distribution of bootstrap estimates for each decay time. The standard error and 95% confidence interval were taken from the standard deviation and the 2.5th to 97.5th percentiles of this distribution respectively. This approach makes no assumption about the underlying noise distribution and naturally captures asymmetric uncertainties.

3.1. Results for Self-Quenching Samples

As shown in Figure 3, the self-quenching data shows a clear linear increase in k_{app} with increasing ZnTPP concentration. The gradient yields a self-quenching rate constant of $k_q = (1.17 \pm 0.03) \times 10^{-4} \text{ L mmol}^{-1} \text{ ns}^{-1}$, with an intrinsic decay rate of $k_1 = (1.093 \pm 0.002) \times 10^{-3} \text{ ns}^{-1}$, corresponding to an intrinsic lifetime of $\tau_0 = 915 \text{ ns}$. This is in agreement with literature values for ZnTPP triplet lifetimes in polar solvents, typically reported in the range of 800 to 1000 ns [1]. The consistent linear increase suggests that bimolecular encounters between excited-state and ground-state ZnTPP molecules provide an additional non-radiative decay pathway, corresponding to the $k_3 C_G$ term in the rate equation.

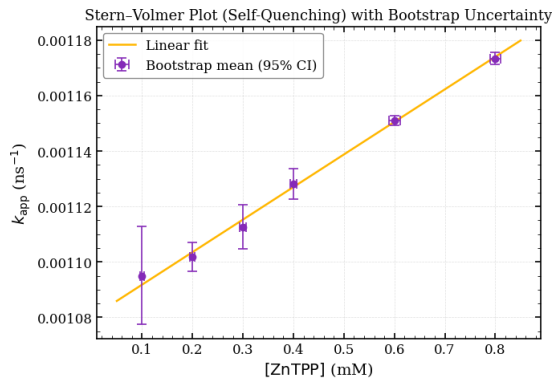


FIG. 3. Dependence of the apparent rate constant k_{app} on ZnTPP concentration for self-quenching samples. A linear fit yields a reduced χ^2 of 0.123.

The linearity further indicates that the quenching mechanism is collisional rather than static, as static quenching would typically show curvature at higher concentrations [4]. However, the low reduced χ^2 value of 0.123 suggests that the bootstrap uncertainties on individual decay times are conservative, leading to an overestimation of errors despite the well-behaved Stern-Volmer kinetics.

3.2. Results for C_{70} Quenched Samples

As shown in Figure 4, the C_{70} quenched samples show a substantial increase in k_{app} with quencher concentration compared to the self-quenching case, yielding $k_q = (1.00 \pm 0.05) \times 10^{-3} \text{ L mmol}^{-1} \text{ ns}^{-1}$ and $k_{eff} = 1.27 \times 10^{-3} \text{ ns}^{-1}$. This is in good agreement with the literature value of $9.8 \times 10^{-4} \text{ L mmol}^{-1} \text{ ns}^{-1}$ reported by Fujitsuka et al [1]. The bimolecular quenching rate constant is approximately 8.5 times larger than the self-quenching value, showing that C_{70} is a significantly more efficient quencher of the ZnTPP triplet state.

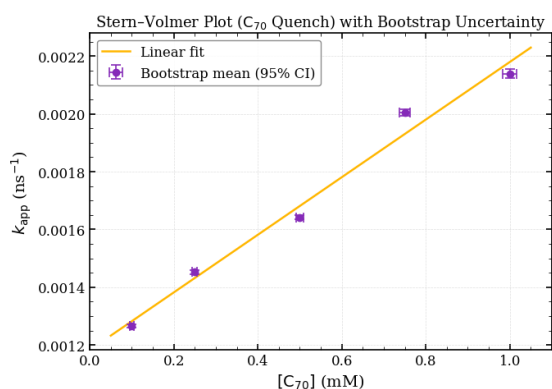


FIG. 4. Dependence of the apparent rate constant k_{app} on C_{70} concentration for C_{70} quenched samples. A linear fit yields a reduced χ^2 of 35.348.

This is consistent with the longer triplet lifetime and stronger visible absorption of C_{70} , both of which increase the probability of bimolecular collisional encounters. The elevated reduced χ^2 value reflects the substantial scatter in the individual k_{app} values beyond what the bootstrap errors accounted for, likely

arising from systematic experimental variability at higher quencher concentrations.

3.3. Results for C_{60} Quenched Samples

As shown in Figure 5, the C_{60} quenched samples display a similar linear trend to that of C_{70} , yielding $k_q = (7.3 \pm 0.4) \times 10^{-4} \text{ L mmol}^{-1} \text{ ns}^{-1}$ and $k_{eff} = 1.23 \times 10^{-3} \text{ ns}^{-1}$. This quenching rate constant is approximately 1.4 times lower than that of C_{70} , consistent with its shorter triplet lifetime and weaker visible absorption, which reduce the probability of collisional energy transfer to the ZnTPP triplet state. This is in agreement with the trend reported in the literature [1]. The reduced χ^2 of 46.8 indicates a similar fitting issue to the C_{70} case, likely reflecting a systematic scatter rather than a failure of the Stern-Volmer model. Repeating the measurements with more concentrations and larger bootstrap resamples would help reduce this scatter and improve the reliability of the fitted parameters.

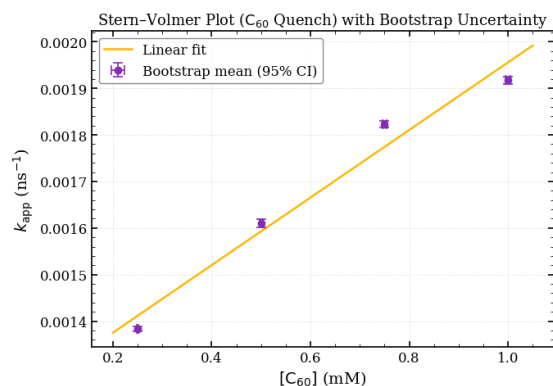


FIG. 5. Dependence of the apparent rate constant k_{app} on C_{60} concentration for C_{60} quenched samples. A linear fit yields a reduced χ^2 of 46.8.

4. CONCLUSION

In this experiment, the triplet-state decay kinetics of ZnTPP were investigated using nanosecond time-resolved absorption spectroscopy with a pump-probe setup. The bimolecular quenching rate constants were found to be $k_q = (1.17 \pm 0.03) \times 10^{-4} \text{ L mmol}^{-1} \text{ ns}^{-1}$ for self-quenching, $k_q = (1.00 \pm 0.05) \times 10^{-3} \text{ L mmol}^{-1} \text{ ns}^{-1}$ for C_{70} , and $k_q = (7.3 \pm 0.4) \times 10^{-4} \text{ L mmol}^{-1} \text{ ns}^{-1}$ for C_{60} . The higher quenching rate constant of C_{70} relative to C_{60} is consistent with its longer triplet lifetime and stronger visible absorption. These results demonstrate that fullerene quenchers provide an effective means of probing intermolecular energy transfer in photoexcited systems.

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